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(54) **STATISTICAL PATTERN ANALYSIS METHODS OF PARTIAL DISCHARGE MEASUREMENTS IN HIGH VOLTAGE INSULATION**

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(52) **U.S. Cl.** **702/58**; 324/536
(58) **Field of Search** 702/57–59, 81–84, 702/179–185; 706/21, 47, 52; 324/512, 522, 536, 551, 772

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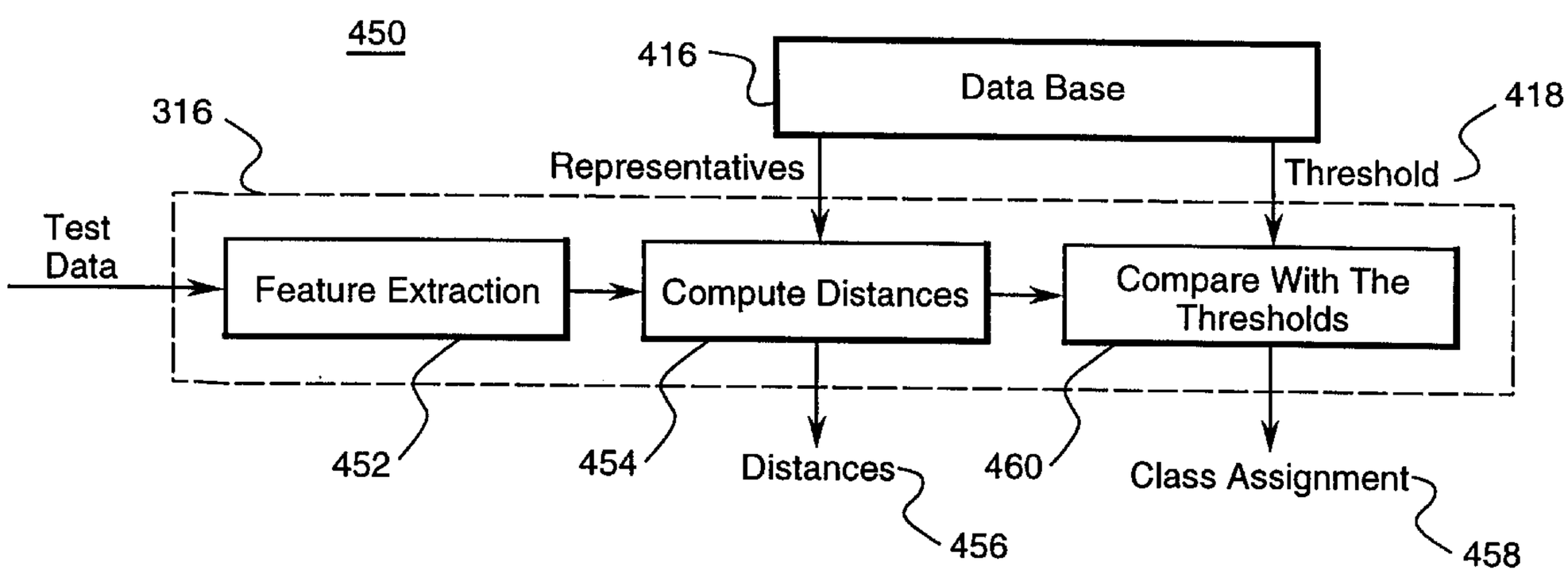
(57) **ABSTRACT**

Statistical methods of partial discharge analysis utilizing histogram similarity measures are provided by quality assessment and condition monitoring methods of evaluating high voltage electrical insulation. The quality assessment method is utilized to evaluate the quality of insulation within the electrical equipment. The condition monitoring method is utilized during normal operation of the equipment to identify degradation in the insulation and to predict catastrophic insulation failures.

10 Claims, 4 Drawing Sheets



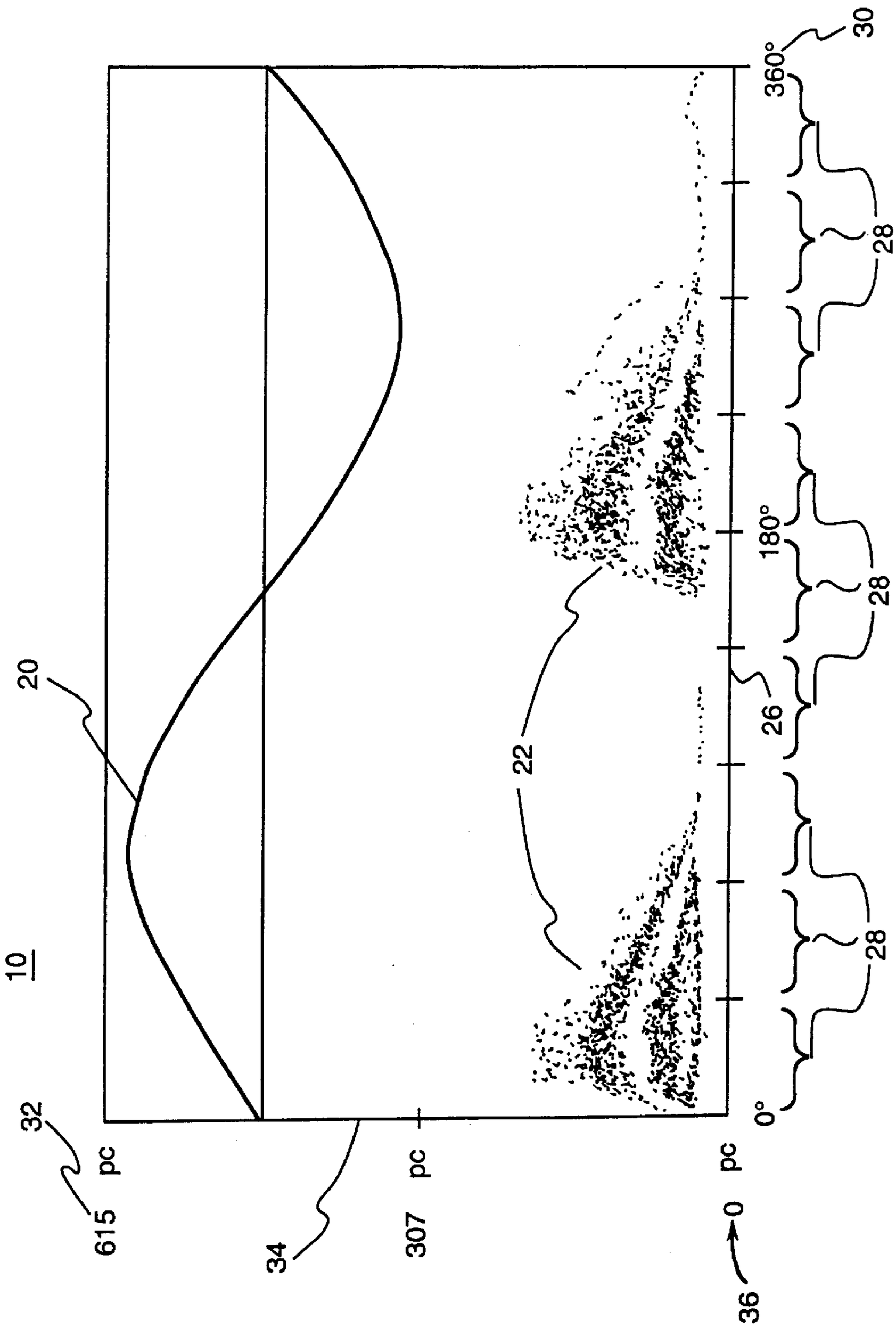
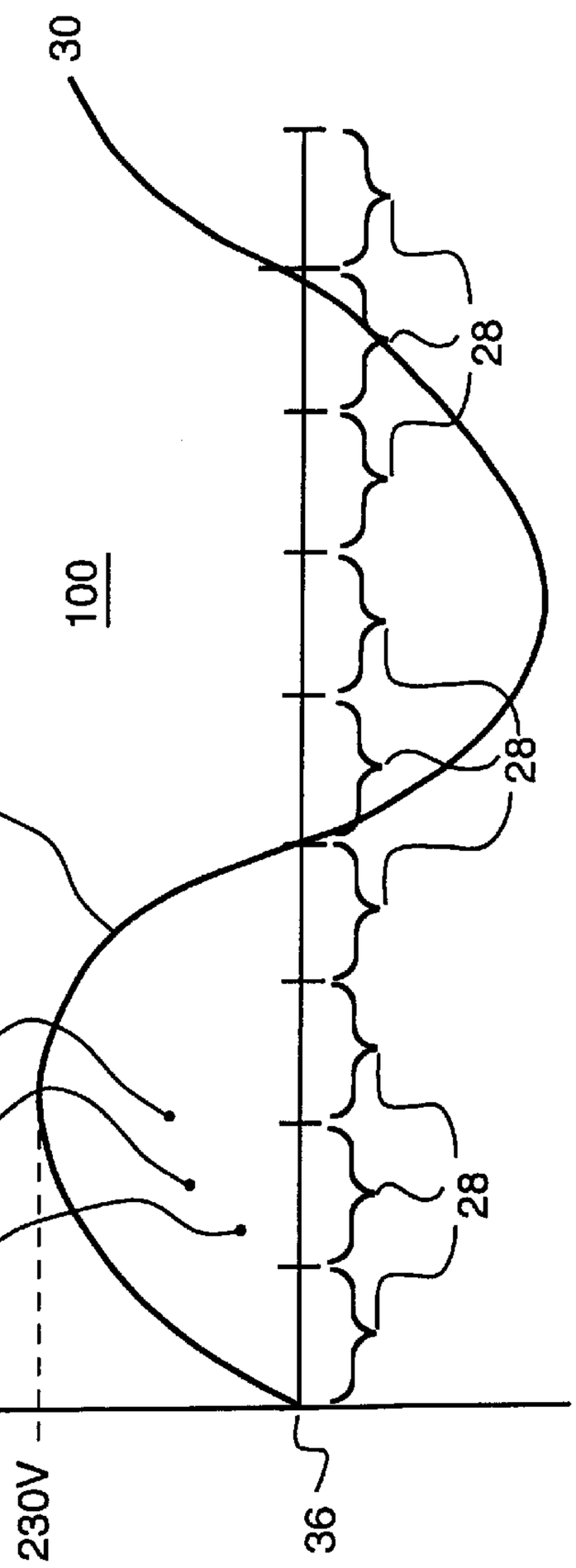


FIG. 1


100
30
20
64
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68
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28
230V
36

FIG. 2

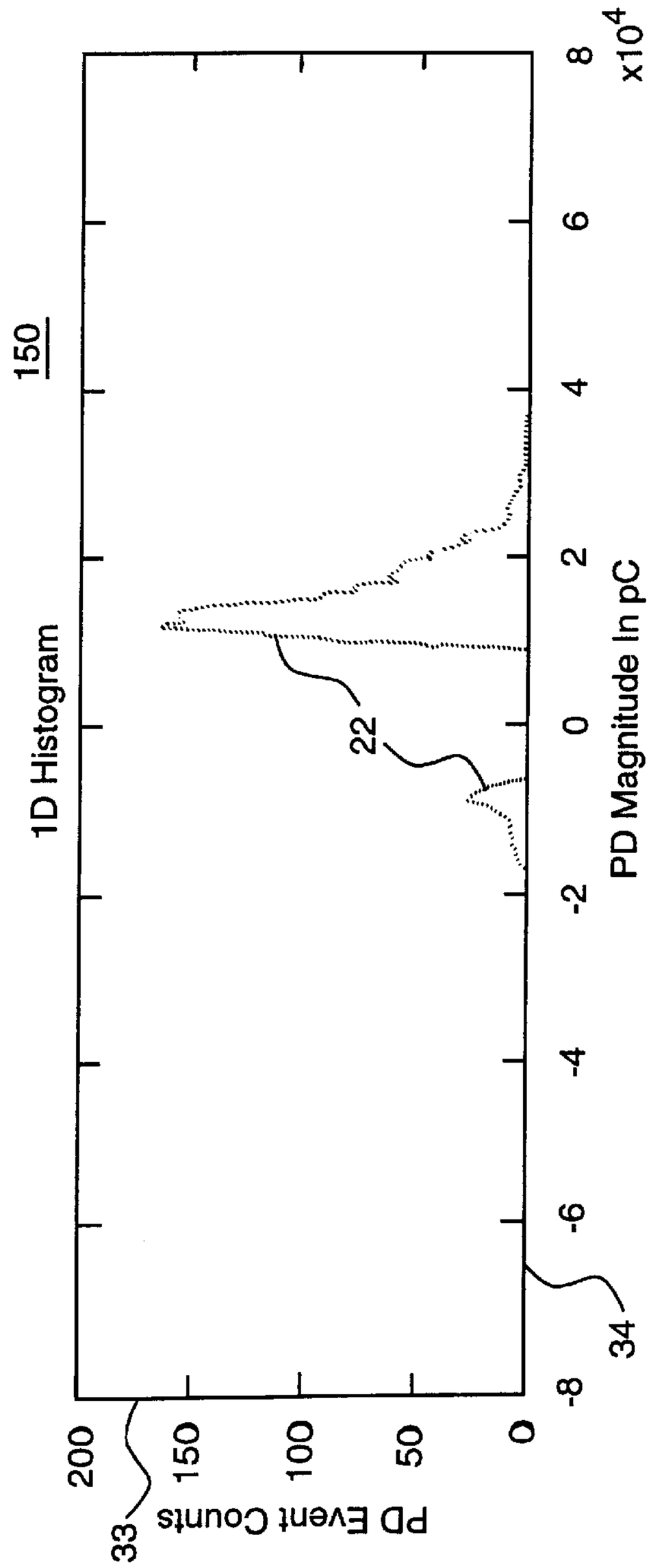
150
1D Histogram
PD Event Counts
200
150
100
50
0
33
22
34
-8
-6
-4
-2
0
2
4
6
8
x10^4
PD Magnitude In pC

FIG. 3

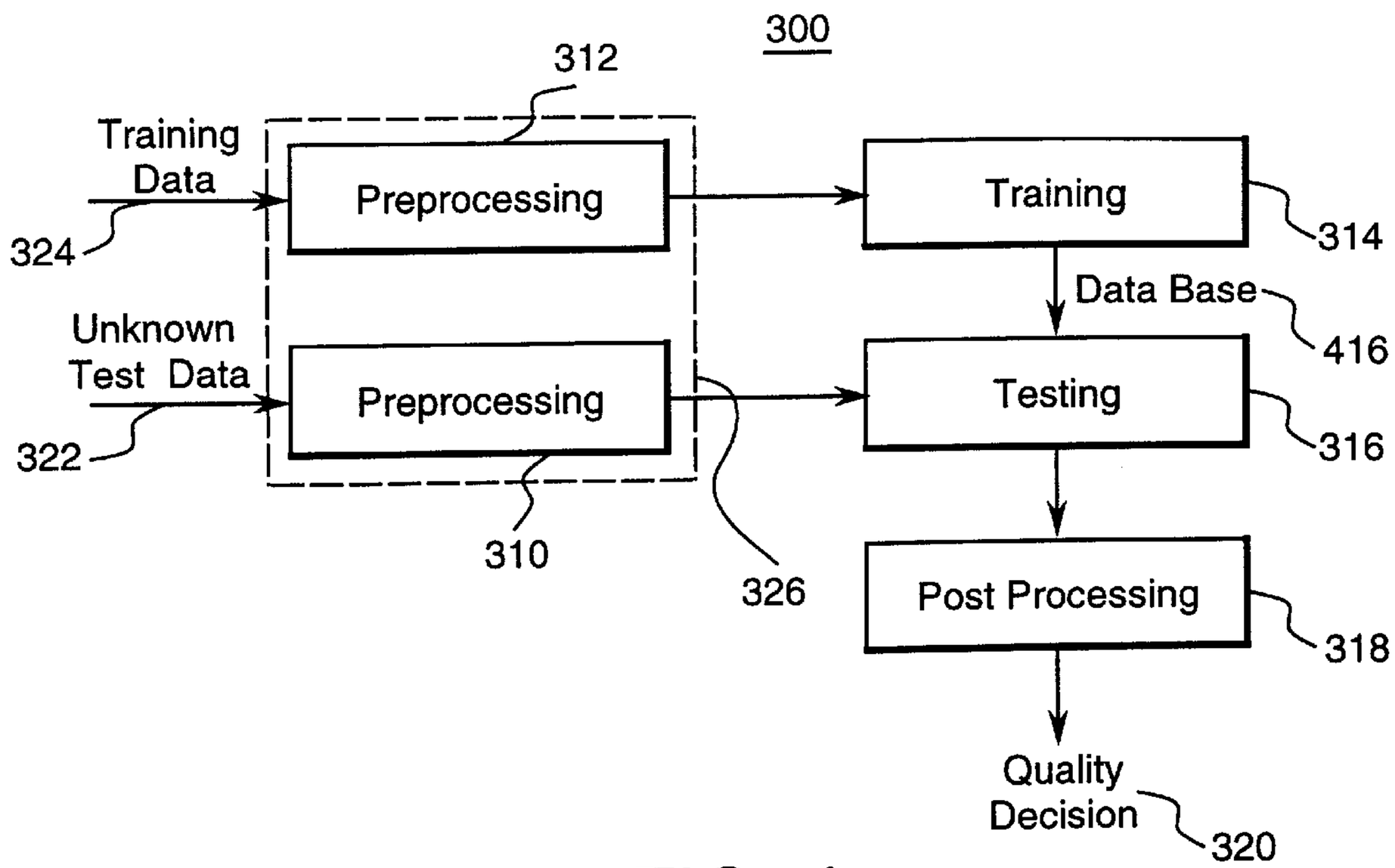
300
312
Training Data
324
Preprocessing
Training
314
Data Base
416
Unknown Test Data
322
Preprocessing
Testing
316
310
326
Post Processing
318
Quality Decision
320

FIG. 4

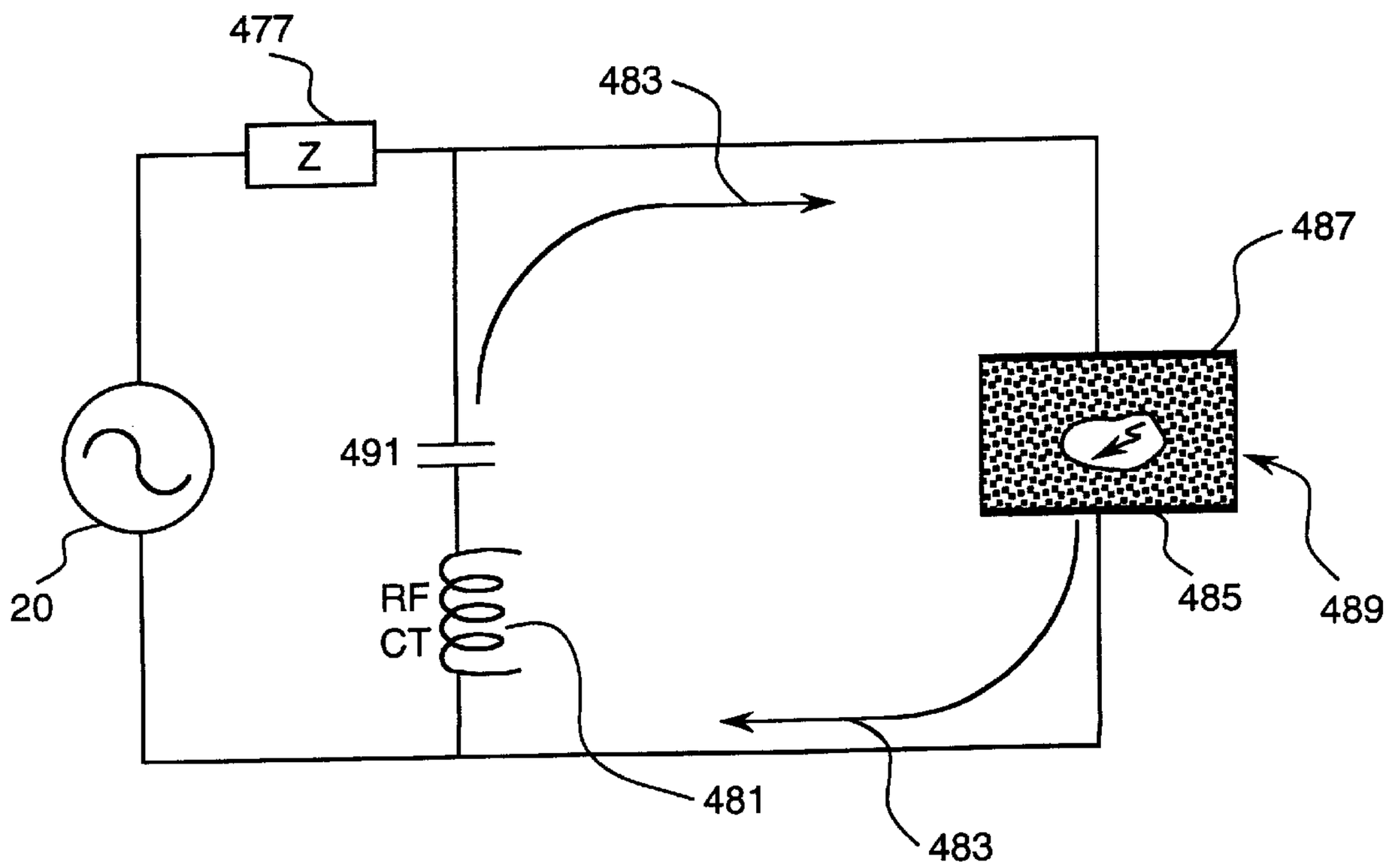
477
Z
483
487
491
20
RF
CT
485
489
481
483

FIG. 7

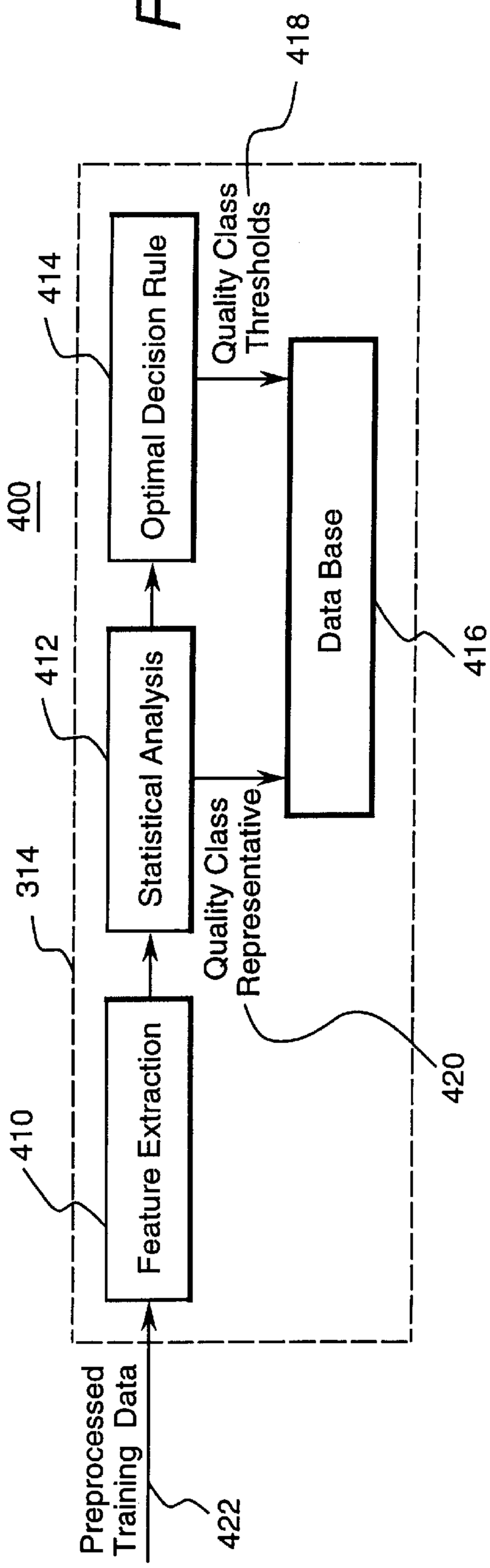
400
Preprocessed Training Data
422
410
Feature Extraction
314
412
Statistical Analysis
414
Optimal Decision Rule
Quality Class Thresholds
418
Quality Class Representative
420
Data Base
416

FIG. 5

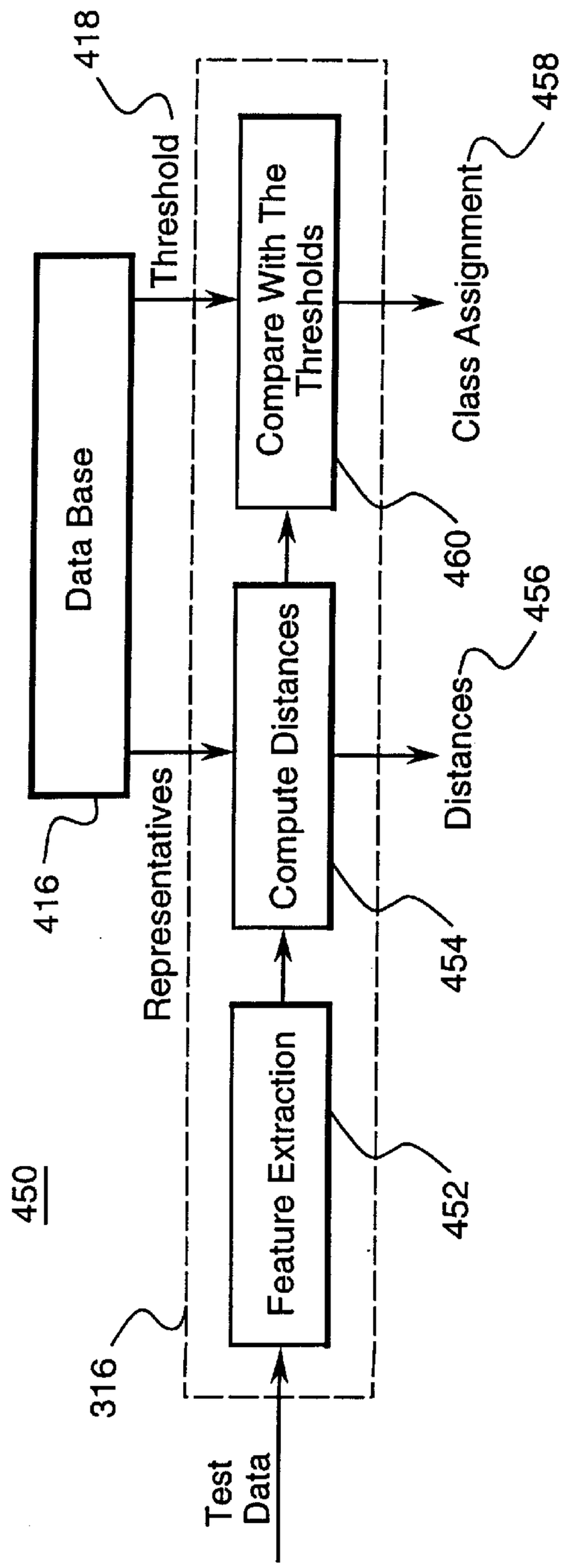
450
Test Data
316
Feature Extraction
452
Compute Distances
454
Distances
456
Compare With The Thresholds
460
Class Assignment
458
Representatives
416
Threshold
418
Data Base

FIG. 6

STATISTICAL PATTERN ANALYSIS METHODS OF PARTIAL DISCHARGE MEASUREMENTS IN HIGH VOLTAGE INSULATION

This application is a division of application Ser. No. 08/837,932, filed Apr. 11, 1997 now U.S. Pat. No. 6,088, 658, which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

This invention relates to quality assessment and condition monitoring of electrical insulation and more particularly to statistical pattern analysis and diagnostics of partial discharge measurements of high voltage insulation.

Partial discharge (PD) analysis has been established as a useful diagnostic tool to asses high voltage insulation systems for integrity and design deficiencies. Interpretation of the PD patterns can reveal the source and the reason for the PD pattern occurrence, and therefore, has been used as a condition monitoring and quality control tool by the manufacturing industry. Analysis of PD patterns for high voltage electrical equipment has often times been an heuristic process based on empirical reasoning and anecdotal information. This is particularly true for insulation systems encountered in high voltage rotating machinery. Typically high voltage insulation is a heterogeneous composite comprising tape, mica flakes, and resin. No insulation system is perfect and there is a statistical distribution of voids and other defects throughout the insulation system. This void distribution results in a baseline level of PD activity for all insulation systems. The associated discharge phenomena are often complicated, multi-faceted events. Discerning between "normal" and defective insulation systems often require more than simple trending of discharge levels.

For many years, the use of PD results for insulation was impracticable because of the difficult data manipulation problems presented. It is thus desirable to identify defective high voltage electrical equipment by the analysis of PD events in an effective and economic manner.

Typically, users of expensive high voltage electrical equipment incur extraordinary expenses when the equipment unexpectedly fails. The ability to predict failures of this equipment would enable the equipment user to utilize condition-based maintenance techniques to avert such unexpected failures and associated high costs. Additionally, scheduled maintenance plans cause users to incur unnecessary costs when equipment is found to be functioning satisfactorily after the scheduled maintenance. Therefore, there exists a need to monitor high voltage insulation when the high voltage equipment is in operation to predict when a catastrophic defect will occur so as to avoid excessive damage and unexpected and costly repair of the high voltage electrical equipment caused by the defect.

There exists a need to identify when a defect exits in insulation before it is placed in the high voltage equipment by assessing the quality of the insulation. It is also desirable to identify the type of defect in the insulation after it is detected.

SUMMARY OF THE INVENTION

The present invention addresses the foregoing needs by providing a system for evaluation of the insulation system of high voltage equipment, including quality assessment and condition monitoring.

The quality assessment method includes extracting representative quality class thresholds from partial discharge

(PD) measurements of known insulation quality during a training stage. These quality class representatives and quality class thresholds from the baseline are used for comparison to analysis features of partial discharge test data to assess the quality of undetermined insulation. Histogram similarity measures are utilized to determine the representatives of the quality class thresholds of each insulation quality type.

The condition monitoring method includes collecting periodic PD measurements of insulation during the normal operation of high voltage equipment. Trend analysis is employed to identify any deviation from an expected baseline of high voltage electrical equipment.

BRIEF DESCRIPTION OF THE DRAWINGS

The features of the invention believed to be novel are set forth with particularity in the appended claims. The invention itself, however, both as to organization and method of operation, together with further objects and advantages thereof, may best be understood by reference to the following description in conjunction with the accompanying drawings in which like characters represent like parts throughout the drawings, and in which:

FIG. **1** is an illustration of a two dimensional phase resolved PD plot utilized in this invention.

FIG. **2** is an illustration of the voltage cycle over which a PD measurement is taken.

FIG. **3** is an illustration of a one dimensional PD plot utilized in this invention.

FIG. **4** is a illustration the major steps of the quality assessment method of this invention.

FIG. **5** is an illustration of the major steps of the training step of the quality assessment method.

FIG. **6** is an illustration of the major steps of the testing step of the quality assessment method.

FIG. **7** is an illustration of the test configuration for making PD measurements during the quality assessment method.

DETAILED DESCRIPTION OF THE INVENTION

In this Specification, two data analysis methods are presented for intelligent characterization and interpretation of partial discharge PD patterns including: a statistical classification method utilizing histogram based analysis to analyze the quality of an insulation system; and a statistical trending method utilizing histogram analysis for the condition monitoring of insulation systems.

High voltage equipment includes for example, high voltage generators, motors, transformers, and transmission lines.

As illustrated in FIG. **1**, the PD measurement is presented in a two dimensional PD plot **10** in which the horizontal axis **26** is delineated by a predetermined number of phase windows **28** and the vertical axis **34** is delineated by the magnitude of each PD event set **22** measured in pico-farads **32**. Two dimensional PD plot **10** is phase resolved, that is, PD event set **22** is taken at predetermined intervals or phase windows **28** in correspondence with a single voltage cycle **20**.

During the quality assessment method, a set of PD reference measurement data **324** (FIG. **4**) are acquired from high voltage equipment having various known defects and from equipment having no defects. A representative PD measurement is extracted from each of these reference

measurements and a respective histogram threshold value that is utilized to identify defects and defect types is determined by computing the histogram similarity distance between the undetermined insulation and the representative PD measurements within the reference PD data set. The respective histogram threshold value **418** and the representative PD measurement are then stored in a data base **416** to be used as baselines for insulation of undetermined quality. When undetermined high voltage insulation is tested, PD measurements **322** are acquired and compared to data from reference PD data **324**, and a statistical distance difference is calculated. If the statistical distance is less than histogram minimum threshold value **418** the high voltage insulation is identified as being within normal operating parameters, hereinafter referred to as "healthy." Otherwise, it is identified as defective high voltage insulation.

During condition monitoring, analysis test data **322** are acquired periodically and consecutive measurements are compared using a histogram similarity measure. If there is a significant deviation in the histogram similarity measure as compared to the measurements acquired in the past, the algorithm generates a indication in correspondence with the degradation in the insulation system. The amount of the deviation in the newly acquired histogram similarity measure may be correlated with the severity of the degradation in the insulation system.

1(a) Histogram Type Measures

In both the quality assessment and condition monitoring methods histogram similarity measures are utilized. PD measurements **22** are grouped by phase window **28** with respect to one cycle of source voltage **20** in phase resolved PD measurement data. For example, one cycle of source voltage **20** is a single cycle of a sixty hertz signal as illustrated in FIG. **2**. The single cycle of source voltage **20** is divided into the preselected number of phase windows **28**. For example,. when two-hundred-fifty-six phase windows are selected, each phase window **28** is represented by approximately 1.4 degrees of source voltage phase. Each phase window **28** collects respective PD measurement data **64**, **66**, and **68** for about thirty seconds of voltage cycles corresponding with its relative position within source voltage **20**. This data forms the basis of a histogram footprint.

A one dimensional histogram has as its horizontal axis the PD magnitude in pico-farads **34** and the vertical axis as the PD event count **33** as illustrated in FIG. **3**. PD event set **22** is plotted in the one dimensional histogram. Phase resolved PD plot **10** is illustrated as a collection of these one dimensional histograms.

Alternatively, only the total number of occurrences need be evaluated for each phase window **28** or PD magnitude **34**, i.e., the sum of individual PD events **22** for each phase window **28** or for each PD magnitude **34**. The histogram obtained by the sum of the individual PD events **22** for each phase window **28** is the phase marginal histogram or X-marginal histogram, and the histogram obtained by the sum of the individual PD events for each PD magnitude is the magnitude marginal histogram or Y marginal histogram.

1(b) Histogram Similarity Measures

The following histogram similarity measures are utilized by way of example and not limitation, to process the previously described histograms: (1) Sample correlation, (2) Kolmogorov-Simimov distance, and (3) Chi-square test.

To simplify the discussion, these measures shall be described for one dimensional histogram **150**. The two dimensional description is a simple extension of several one dimensional histograms.

The normalized joint probability mass function of phase windows **28** and PD magnitudes is computed by dividing the histogram into the total unit area. The Probability Mass Function or Normalized Histogram is defined by equation 1 below.

$$P(i) = \frac{h(i)}{A},\ i = 1, \ldots, N, \tag{1}$$

where h(i) is the number of events whose intensity is "i", "A" is the total number of events in the observation set, and "N" is the number of phase windows **28** over a single voltage cycle.

The Cross Correlation value "cr" as defined in equation 2 below:

$$cr = \frac{\sum_{i=1}^{N}(P_1(i)-\mu_1)(P_2(i)-\mu_2)}{\sqrt{\sum_{i=1}^{N}(P_1(i)-\mu_1)^2 \sum_{i=1}^{N}(P_2(i)-\mu_2)^2}}, \tag{2}$$

where μ_i is the mean value of probability mass function P(i).

The Kolmogorov-Simimov Distance is calculated using the Cumulative Distribution function, the Kolmogorov Simirnov distance 1 function, and the Kolmogorov Simirnov distance 2 function.

The Cumulative Distribution Function is defined by equation 3 below:

$$H(k) = \sum_{i=1}^{k} P(i), \tag{3}$$

where $H_i(k)$ is the probability that the intensity "i" is less than or equal to k.

The Kolmogorov Simimov distance 1 between two cumulative distributions is defined by equation 4 below:

$$KS1 = \max_k |H_1(k) - H_2(k)|, \tag{4}$$

The Kolmogorov Simirnov distance 2 is defined by equation 5 below:

$$KS2 = \sum_{k=1}^{N} |H_1(k) - H_2(k)|. \tag{5}$$

Alternatively, the Chi-Square Test is utilized to calculate the normalized histogram. The Chi-square statistic is described as follows:

$$\chi^2 = \sum_{i=1}^{N} \frac{(h(i) - AP(i))}{AP(i)} \sum_{i=1}^{N} p_i = 1.\ N_i\ i = 1, \ldots, N \tag{6}$$

where P(i) is a known probability mass, and h(i)' is the number of events at intensity "i", where

$$\sum_{i=1}^{N} h(i) = A.$$

It is desirable to know whether h(i)'s are drawn from the population represented by the P(i)s. If the h(i)'s are drawn

from the population represented by P(i)s, then as $A \rightarrow \infty$, each term in the summation of the equation (6) is the square of a Gaussian random variable with zero mean and unit variance. Therefore, x^2 is a random variable with Chi-square distribution. A large value of x^2, for example, $x^2 >> N$, indicates that it is rather unlikely that the h(i)'s are drawn from the population represented by P(i) because $Pr(x^2>C|N)$ is very small.

$Pr(x^2>C|N)$ is the probability that the Chi-square statistic is at least "C", given that there are "N" phase windows **28**, alternatively called bins, in the histogram and that h(i)'s are drawn from the population represented by P(i)s. The value of $Pr(x^2>C|N)$ is computed by the incomplete gamma function below:

$$Pr(\chi^2 > C \mid N) = \frac{\int_{C/2}^{\infty} e^{-x} x^{N/2-1}}{\int_{C/2}^{\infty} e^{-x} x^{N/2-1}}. \tag{7}$$

The Chi-Square test for two histograms is described next. Since the precise probability mass functions of PD event set **22** is not ascertainable, what is implemented is a modified Chi-square test which checks whether two measurements $h_1(i)$, i=1, . . . , N, and $h_2(i)$, i=1, . . . , N are drawn from the same population by a predetermined probability distribution function. In this case the Chi-square statistic is given by:

$$\chi^2 = \sum_{i=1}^{N} \frac{\left(\sqrt{A_2/A_1}\, h_1(i) - \sqrt{A_1/A_2}\, h_2(i)\right)}{h_1(i) + h_2(i)} \tag{8}$$

$$\text{where } \sum_{i=1}^{N} h_1(i) = A_1 \text{ and } \sum_{i=1}^{N} h_2(i) = A_2.$$

In this implementation, $h_1(i)$ is the first histogram, and $h_2(i)$ is the second histogram of PD measurements. Note that if $h_1(i)=h_2(i)=0$ for some phase window "i", then the corresponding term is dropped in the summation in equation (8).

2. Quality Assessment

The quality assessment method is a "supervised" approach, that is, the type and the number of reference quality classes, against which the insulation of high voltage equipment is tested, are predetermined. It is utilized for detection, as well as the identification of reference quality classes. Detection involves automatic decision making to determine whether high voltage equipment has "healthy" or "defective" insulation. Identification, on the other hand, involves defect detection, as well as, classification of the defect to one of the known quality classes.

As more information is provided to the system, better inference of the quality of the insulation can be achieved. For instance, for defect detection, only a set of PD measurements representing a "healthy" insulation system is needed. For quality class identification, however, PD measurements from each quality class are needed. The following list, for example, is a group of quality classes that are detected including: suspended metal particles in insulation; armor degradation; conductor and insulator delamination; insulator/insulator delamination; and insulator/armor delamination. A list of these failures are illustrated in Table 1. The outcome of such a quality assessment utilizing the present invention generates one of these three associated quality class group indications, "healthy" insulation, "defective" insulation of known quality class, and "defective" insulation of undetermined quality class. As more quality classes are programmed into the reference set of detect types the identification capability of the system improves.

The quality assessment method comprises four stages including: pre-processing **326**, reference **314**, testing **316**, and post-processing **318** as illustrated in the flow diagram in FIG. **3**. In pre-processing stage **312**, PD reference data **324** may be subjected to signal conditioning methods, such as gain normalization, phase synchronization and excitation noise suppression. In reference stage **314**, the characteristics of the defected and good insulation are learned and memorized from reference data **324**. The learning process involves extraction of the characteristics or quality class features from reference data **324** and statistical analysis of the quality class features. Statistical analysis of the quality class features leads to a set of quality class representatives **420** for the quality class features and an optimal decision making rule **414** as illustrated in FIG. **5**. Using the reference data and the decision rule, quality class thresholds **418** is calculated for each quality class feature. Finally, a respective quality class representative **420** of each quality class along with a respective quality class thresholds **418** is stored in data base **416** for use during testing stage **316**. In testing stage **316**, respective quality class features **452** are extracted from a each test bar and a statistical distance between quality class representative **420** of each respective quality class and the quality class features is computed. The resulting respective distance **456** is compared with respective quality class thresholds **418** in data base **416** to assign the test measurement to one of the quality classes stored in data base **416**. In post-processing stage **318** results from multiple PD measurements are combined to increase the confidence level and a probability is associated with each decision to quantify the accuracy of the process. Each of the four quality assessment stages is discussed in more detail below.

2.(a) Pre-Processing

In prepossessing stage **326**, reference data **324** and analysis test data **322** may be conditioned in any one of the following ways or in combination with multiple sets of the following ways: a) to convert from bipolar data to uni-polar data; b) to phase synchronize with supply voltage **20**; c) to be normalized with respect to a gain factor; and d) digital filtering may be utilized to suppress the excitation noise in reference data **324** and analysis test data **322**.

2.(b) Training

The flow diagram of the training process **314** is illustrated in FIG. **5**. Low count high magnitude PD events contain more discriminative information than the low magnitude high count PD events. In one embodiment of the feature extraction process, the logarithm of the PD event counts is calculated. In addition the effect of high count low magnitude events may be suppressed by setting a predetermined lower-level PD magnitude threshold discarding data below this value. Typically, the lower-level PD magnitude threshold is the bottom ten percent of the PD magnitude range.

Respective quality class representatives **420** from each quality class is extracted by averaging reference data **324**. Expressed mathematically let R_i denote the quality class representative **420** of the quality class i and P_i^j be the "jth" member of the quality class "i." Next, an intra quality class distance between its quality class representative **420**, R_i, and each of its member, R_i^j, in reference data set **324** is calculated by using the histogram similarity measures introduced above.

$$D_i^j = d(P_i^j, R_i) \tag{9}$$

where d is the histogram similarity measure. Note that in the case of 1D histogram similarity measures, the distance function is a vector in which each entry is the distance between the member and reference 1D histograms of respec-

tive phase window **28**. To clarify this difference, vector notation for the 1D histograms is introduced:

$$D_i^j(\omega)=d(P_i^j(\omega),R_i(\omega)),\ \omega=1,\ldots,L. \tag{10}$$

ω corresponds to the phase window **28**, and the value L was selected based on the number of phase windows **28** over a single cycle of source voltage **20** as discussed above.

In order to obtain an optimal radius for each quality class, the sample mean and standard deviation of the distances within each quality class are calculated. For 2D histograms the intra quality class mean radius and standard deviation are given as follows:

$$\overline{D}_i = \frac{1}{N_i}\sum_{j=1}^{N_i} D_i^j \tag{11}$$

$$\sigma_i = \sqrt{\frac{1}{N_i}\sum_{j=1}^{N_i}(D_i^j - \overline{D}_i)^2} \tag{12}$$

where N_i is the number of samples available for the quality class "i." For the 1D phase resolved histograms the intra-class mean radius and standard deviation are given by

$$\overline{D}_i(\omega) = \frac{1}{N_i}\sum_{j=1}^{N_i} D_i^j(\omega) \tag{13}$$

$$\sigma_i(\omega) = \sqrt{\frac{1}{N_i}\sum_{j=1}^{N_i}(D_i^j(\omega) - \overline{D}_i(\omega))^2},\ \omega = 1,\ldots,L. \tag{14}$$

Next, an α unit standard deviation tolerance is allowed for each quality class radius. Note that in the case of Gaussian distribution of the intra-class distances, α is typically chosen to be two to provide a 99% confidence interval. The quality class radius, for the respective 2D and 1D histograms are given as follows:

$$\tau_i=\bar{D}_i+\alpha\sigma_i \tag{15}$$

$$\tau_j = \sum_{\omega=1}^{256}\overline{D}_i(\omega) + \alpha\sigma_i(\omega),\ \omega = 1,\ldots,L. \tag{16}$$

Finally, for each quality class $\{R_i, \tau_i\}$ are stored in the data base as quality class representative **420** and the quality class threshold **418** of each quality class.

2.(c) Testing

The major steps of testing stage **316** is illustrated in FIG. **6**. Analysis test data **322** is utilized in the following steps. First, analysis test data **322** are subjected to the steps discussed in pre-processing stage **312** prior to extracting the relevant features. Next, the distance between the analysis feature, T, and the references of each quality class R_i are computed using one of the histogram similarity measures described above. For the 2D histogram, the distance is given by a scalar quantity (equation 17) while for the 1D histogram similarity measures, it is given by a vector quantity (equation 18):

$$\Delta_i=d(T,R_i)\ i=1,\ldots,C \tag{17}$$

$$\Omega_i(\omega)=d(T(\omega),R_i(\omega)),\ \omega=1,\ldots,L,\ i=1,\ldots,C \tag{18}$$

where "C" is the number of quality classes. In order to compare the vector quantity identified in equation (18), with quality class threshold **418**, a cumulative distance is computed by summing the quantity in equation (18) over the phase windows **28**:

$$\Delta_i = \sum_{\omega=1}^{L}\Omega_i(\omega) \tag{19}$$

Next, the respective distance between the analysis features and quality class representative **420** of each quality class, Δ_i, are calculated. If the distance is greater than all quality class thresholds **418**, the test measurement is tagged as "defective." However, the quality class is not one of the types for which the algorithm is trained to recognize. If the distance is less than one or more quality class thresholds **418**, the test measurement is assigned to the quality class to which the distance between the associated reference and analysis feature is minimum. If the final assignment is the quality class of healthy insulation, the measurement is tagged as "healthy."

To improve the accuracy of this quality class assignment, testing stage **316** is repeated for multiple sets of analysis test data **322**. Decisions obtained for each set of analysis test data **322** and distances of each feature to each quality class representative **420** are then subjected to post-processing **318** to finalize the decision of each undetermined insulation system.

2.(d) Post-Processing

In post-processing stage **318**, there are two major tasks: first, the decisions in testing stage **316** are combined; second, a probability is associated to each decision to reflect the accuracy of the quality classification.

The distance of the analysis feature to a quality class as illustrated in equation (17) is quantified in terms of unit standard deviation of the quality class. Relative comparison of these distances determines the accuracy of the final decision. The distances are mapped to a probability distribution function in order to compare these decisions. If the distance between the analysis feature and the quality class is zero, or significantly larger, such as α^2, The classification decision is about 100% accurate. In the case of zero distance, analysis test data set **322** belongs to the quality class with probability 1. In the case of α^2 or larger distances, the probability that analysis test data set **322** belongs to the quality class is zero. The difficulty arises when assigning a probability to the distances that are between zero and α^2. To resolve this issue, equation (20) is used between the distance and probability spaces.

$$f(\Delta) = \begin{cases} \left(1-\frac{\Delta}{\gamma}\right)^{-\beta} & 0 \le \Delta < \gamma \\ 0 & \Delta \ge \gamma \end{cases} \tag{20}$$

Note that when the distance is zero, i.e., Δ=0, the probability, f(Δ), is 1, and when the distance is greater or equal to γ, the probability is zero.

Two parameters in equation (20) are user-controlled, namely γ and β. The parameter γ controls the threshold beyond which the probability that a measurement coming from a particular quality class has distance γ or larger. This distance can be set to an order magnitude larger than the threshold α, such as α^2 unit standard deviation. The parameter β controls the confidence in quality class identification power of the testing scheme. This parameter is preselected so that 80% or more values should be correctly classified. If a value is chosen at the 80% quality class identification level, β is given as follows:

$$\beta==In0.8/In(1-\alpha/\gamma) \tag{21}$$

Or more generally, choose β so that $f(\alpha)$ is equal to the confidence level defined as a probability percentage. For example, when $f(\Delta)$, is selected to be 80% γ, may be selected to be four and β is calculated to be −0.322. These parameters enable a quality class likelihood to be assigned each analysis test data set **322**. Other values for equation (20) may also be selected.

While probability mapping functions defined in equations (20) and (21) are utilized in this invention, any technique can be utilized to produce the above results, for example, a computer look-up table, or a set of logical rules defined in a computer program.

Next, the quality class assignments and the associated probability values are combined to get a final quality class assignment and a probability value for analysis test data set **322**. The steps of this process are summarized as follows:

A probability value is assigned to each analysis test data set **322** describing the likelihood of the measurement belonging to each quality class in the database. Let p_i^j, j=1,. . . , M and i=1, . . . , N be the probability that the measurement j belongs to quality class i. where "M" is the total number of measurements.

Next, $\overline{p}_i$ is defined as the probability that the analysis test data set **322** representing a test bar belongs to quality class i with probability $\overline{p}_i$.

$$\overline{p}_i = \frac{1}{M}\sum_{j=1}^{M} p_i^j,\ i = 1, \ldots, C \tag{22}$$

The insulation represented by analysis data set **322** is next assigned to the quality class with the highest overall probability, i.e.,

$$\hat{i}=Argmax/i\ \bar{p}_i \tag{23}$$

where Argmax is the quality class at which $\overline{p}_i$, i=1, . . . N is maximum.

An example of the above described quality assessment method, is presented. Several quality class classes of generator stator bars, hereinafter described as bars or generator bars, were programmed into the statistical model as several respective reference data sets **324**. Next, undetermined generator bars were evaluated based on respective reference data set **324**. The results are described and illustrated below.

Several types of stator bar defects or quality classes were investigated including suspended metallic particles in the insulation, slot armor degradation, corona suppresser degradation (i.e., the external voltage grading), insulation-conductor delaminations, insulator-insulator (bound) delaminations, and "white" insulation. Each of these quality class has implications for either quality assessment or condition monitoring either during production or during service Each bar was subjected to a specific sequential set of events to ensure consistent reliable PD data.

The typical test configuration is illustrated in FIG. **7**. A 60 Hz, corona-free, supply voltage **20** having an associated impedance **477** energizes respective model generator stator bars **489**. Respective stator bars **489** are placed in simulated stator slots **487** and **485** and held firmly. Slots **487** and **485** are made of either aluminum or steel plates cut to fit the slot section of stator bars **489**. A 1.3 nano-farad ceramic coupling capacitor **491** is used as a coupling impedance which allows PD current **483** to pass through current transformer **481** coupled thereto. The bandwidth of data collection is 100,000 to 800,000 Hertz. Data is collected for thirty seconds for each PD pattern. Multiple, sequential patterns are collected for each respective stator bar **489**.

The results of the quality assessment approach for various histogram type similarity measures is next presented as an example of the operation of the quality assessment approach. This example includes comparison of the 2D Chi-square, correlation, and Kolmogorov-Simimov methods of statistically analyzing PD data. The number of test samples and type of reference quality classes used for each quality class is tabulated in Table 1. For quality class matching, both the linear and the logarithmic-scaled PD data were used.

TABLE 1

Number of Testing Samples for each Quality Class

Quality class	Number of Measurements
Good Bar	10
Suppresser	5
Metal Wires	5
No Int Grade	10
Cu Delam	5
Whitebar	5
Therm & VE	4
Arm Control	4
Slot Dischg	4

In this example, there are a total nine quality classes, eight for defects and one for "healthy" insulation. Fifty-two patterns were collected. The respective reference data sets **324** and respective quality class thresholds **418** are determined for each quality class and similarity measure. Quality class thresholds **418** of the correlation method for each quality class are tabulated in Table 2 for both a linear and a logarithmic treatment of the patterns. The linear thresholds are calculated when data from the PD magnitude **34** (FIG. **1**) is a linear scale. The logarithmic threshold are calculated based on a logarithmic PD magnitude **34**.

TABLE 2

Correlation Thresholds for each Quality class

Quality class	Linear Thresholds	LogarithmicThresholds
Good Bar	0.93	0.97
Suppresser	0.79	0.96
Metal Wires	0.95	0.97
No Int Grade	0.87	0.96
Cu Delam	0.98	0.98
Whitebar	0.98	0.98
Therm & VE	0.58	0.98
Arm Control	0.88	0.98
Slot Dischg	0.97	0.95

The "quality class detection" and "quality class identification" power of each histogram similarity measure was determined. Defect detection power is defined as the "probability of detecting" the presence of a defect. "Identification power" is the probability of identifying the quality class correctly. A "false positive" is a quality class identification when no quality class should have been identified. These results are summarized in Table 3.

TABLE 3

Performance of the Testing Stage Using 1D Phase Histograms

Test Type	Defect Detection	False Positive	Defect ID
Chi-squares	100%	70%	78%
Correlation	100%	0%	96%
Kolmogorov-Simirnov 1	100%	0%	96%
Kolmogorov-Simirnov 2	100%	0%	98%

A preferred embodiment of conducting the histogram similarity analysis is to use the 1D phase Kolmogorov-Simimov 2 approach because the it provides the best measure in quality class detection and quality class identification as is illustrated in Table 3. Out of fifty-five pattern sets acquired from the defective bars all of them were correctly identified as defective yielding a 100% defect detection confidence level as is illustrated in Table 4. On the other hand, fifty-five of the fifty-six patterns are correctly identified yielding 98% class identification confidence.

The detailed experimental results for the 1D phase Kolmogorov-Simirnov 2 test are summarized in the confidence matrix illustrated in Table 4. The number in the "ith" row and "jth" column of this table shows how many samples from quality class "i" is classified to quality class "j". For example the row titled "No Int grade" or No internal grading shows that only one sample is miss-classified as quality class "Cu Delam" or Copper delamination degradation while nine samples are correctly classified as the quality class "No Int grade."

TABLE 4

Confidence Matrix for the Kol-Simiornov 2 Test Using 1D Phase Histogram

	Supressor	Metal wire	No Int grade	Cu Delam	White bar	Therm&VE	Slot dischg	Am control	Good bars
Supressor	5								
Metal wire		5							
No Int grad			9	1					
Cu Delam				5					
Whitebar					5				
Therm&VE						4			
Slot dischg							4		
Arm control								4	
Good bars									10

The 1D phase Kolmogorov-Simimov 1 method provides similar results as the 1D phase Kolmogorov-Simimov 1 method (Table 5). It also has 100% quality class detection as indicated in Table 3. The quality class identification, however, is slightly less at 96%. One test method may be more adept at identifying quality classes than another method. In tables 4 and 5 the 1D phase Kolmogorov-Simimov 2 erred in identifying one no grade and the 1D phase Kolmogorov-Simirnov 1 erred in identifying one bared copper/insulation delamination test. These errors are further eliminated by employing multiple statistical distance comparisons using different statistical methods. For example, both the 1D phase Kolmogorov-Simirnov 1 and the 1D phase Kolmogorov-Simirnov 2 can be combined to improve fault identification.

TABLE 5

Confidence Matrix for the Koi-Simiornov 1 Test Using 1D Phase Histogram

	Supr-essor	Metal wire	No Int grade	Cu Delam	White bar	Therm& VE	Slot dischg	Am control	Good bars
Supressor	5								
Metalwire		5							
No Int grade			10						
Cu Delam				5					
Whitebar		1			4				
Therm&VE						4			
Slot dischg	1						3		
Arm control								4	
Good bars									10

3. Condition Monitoring

The objective of a condition monitoring method is to quantify the health of the system to predict catastrophic failures, to avoid excessive damage and unexpected down times, and to allow condition-based maintenance. The condition monitoring method acquires PD measurements periodically and compares them with past PD measurement to detect statistical deviations. Any significant deviation in the most current measurement is identified as an indication of change in the PD activity of the high voltage insulation.

The condition in which the measurements remain statistically unchanged is hereinafter identified in this Specification as a "steady state condition." In a typical generator with a healthy insulation system, a steady state condition lasts about 3 to 6 months. While the measurements in each state remain unchanged, they follow a time dependent trend line over several steady states. The underlying truism in condition monitoring and failure detection system is that the measurements during a pre-catastrophic state do not match to the predicted values obtained from the trend analysis of the earlier states, because catastrophic deterioration is expected to be revealed by the rate of increase in the PD activity. The implementation of the condition monitoring method involves the following steps: 1) preprocessing as discussed above; 2) feature extraction as discussed above; 3) establishing a parametric trend model; 4) estimating the trend parameters; 5) and, predicting the PD trend so as to detect pre-catastrophic failures.

By way of example, and not limitation the "steady state condition" of a high voltage machine is about six months and the data are collected every week on full load operation. Due to external operating conditions and the physical nature of the PD activity, there are variations and noise in day to day PD measurements. To suppress noise, the seven day PD measurement is averaged on a weekly basis. PD measurements are taken at the full load condition of the high voltage electrical equipment. To quantify the daily fluctuations, the distance between the weekly average and the daily measurements is measured. Next, a monthly average is calculated based on four weekly averages. Then weekly fluctuations are calculated based on the monthly average. These fluctuations provide information about the short term behavior of the PD activity. The average value of these fluctuations is expected to be a constant as long as the equipment is not in a pre-catastrophic state. Next, measure the similarity of the weekly and monthly averages to a predetermined previous step.

PD measurements for condition monitoring typically are taken at preselected intervals that depend on the expected rate of insulation deterioration, as such, the weekly and monthly intervals may be any other interval of time.

Statistical analysis used to determine the predicted catastrophic failure is based on condition monitoring is as follows.

Let P_i^m and P_i^w represents the monthly and weekly average PD measurements. Let $D^m(\tau)$ and $D^w(\tau)$ be the distance between τ apart measurements, i.e.,

$$D^m(\tau)=d(P_i^m, P_{i+\tau}^m) \tag{24}$$

and

$$D^w(\tau)=d(P_i^w, P_{i+\tau}^w) \tag{25}$$

where the distance d is one of the histogram similarity measures introduced previously. Typically, the distance between the measurements increases as the time lag τ increases. Therefore, the distance values conform to trend line models of the form:

$$D^m(\tau)=A^m\tau+B^m+\sqrt{\rho^m}N^m(\tau),\ \tau=1, 2, 3, \ldots T \tag{26}$$

and

$$D^w(\tau)=A^w\tau+B^w+\sqrt{\rho^w}N^w(\tau),\ \tau=1, 2, 3, \ldots T \tag{27}$$

where $N^m(\tau)$, $N^w(\tau)$, $\tau=1, 2, 3, \ldots T$ are Gaussian zero mean unit variance white noise processes and p^m and p^w are the variance of $D^m(\tau)$ and $D^w(\tau)$ respectively. "T" is a number representing a predetermined number of PD measurements before the latest PD measurement. For example, "T" may be five steps from the latest measured step. While equations (26) and (27) are expressed as linear models, other models may be utilized such as a hyperbolic model or log linear model of the form presented in equations (28) and (29) respectively.

$$D^m(\tau)=\tau^{\alpha^m}+\beta^m+\sqrt{\delta^m}N^m(\tau),\ \tau=1, 2, 3, \ldots T \tag{28}$$

and

$$D^w(\tau)=\tau^{\alpha^w}+\beta^w+\sqrt{\delta^w}N^w(\tau),\ \tau=1, 2, 3, \ldots T \tag{28}$$

Note that both the parameter A and the parameter α control the rate at which the partial discharge activity changes. While A quantifies a first order polynomial change, α quantifies a change faster than any polynomial but less than an exponential change. The parameter which controls the rate of increase as identified by parameter A and param-

eter α is identified as the "trend rate." The parameter B and the parameter β determine the constant difference between consecutive PD measurements. The parameter B and the parameter β is identified as the "trend constant." The variance of the process as is identified as the "trend variance." The condition during which the trend rate is constant shall be identified as the "steady state condition." The "trend rate" and the "trend variance" of the weekly and monthly averages are expected to increase indicating normal increased PD activity. However, the increase in these parameters is expected to be linear when the insulation system is still relatively healthy. Note that if the time difference τ and the distance $D^w(\tau)$ are statistically independent, the trend rate estimate will be about zero.

The discussion that follows is based on the linear monthly model, however, any of the trend line models identified above may be substituted for the linear monthly model.

3(a) Estimation of the Trend Parameters

The parameters of the model typically is estimated using the linear or log linear least squares method. For example, the linear monthly model provides estimate parameters based on the following formulas:

$$\hat{A}^m = \frac{\sum_{\tau=1}^{T} D^m(\tau)\tau - \frac{1}{T}\sum_{\tau=1}^{T} D^m(\tau)\sum_{\tau=1}^{T}\tau}{\sum_{\tau=1}^{T}\tau^2 - \frac{1}{T}\left(\sum_{\tau=1}^{T}\tau\right)^2} \tag{30}$$

$$\hat{B}^m = \frac{1}{T}\sum_{\tau=1}^{T} D^m(\tau) - \hat{A}^m\frac{1}{T}\sum_{\tau=1}^{T}\tau \tag{31}$$

$$\bar{\rho}^m = \frac{\sum_{\tau=1}^{T}\left(D^m(\tau) - \hat{A}^m\tau - \hat{B}^m\right)}{T-2} \tag{32}$$

3(b) Predicted Partial Discharge Activity

In order to detect an outstanding state, the "trend rate" must be estimated based on past PD measurements and compared with the PD measurement estimated from the new measurements. If the variance of the process is known, the "trend rate" estimate is Gaussian with distribution

$$N\left(\hat{A}^m, \rho^m \Big/ \left(\sum_{\tau=1}^{T}\tau^2 - \left(\frac{1}{T}\sum_{\tau=1}^{T}\tau\right)^2\right)\right). \tag{33}$$

Since the variance of the process is undetermined, the estimate $\hat{A}^m$ is t-distributed with T−2 degrees of freedom. Therefore the (100−2P)% catastrophic confidence interval is given by

$$\left|\hat{A}^m - A^m\right| < t_c \frac{\sqrt{\hat{\rho}^m}}{\left(\sum_{\tau=1}^{T}\tau^2 - \left(\frac{1}{T}\sum_{t=1}^{T}\tau\right)^2\right)} \tag{34}$$

where t_c is such that for the t-distribution with T−2 degrees of freedom, there is P% chance that $t>t_c$. Similarly, for known process variance the "trend constant" estimate is Gaussian with distribution

$$N\left(B^m, \rho^m \sum_{\tau=1}^{T}\tau^2 \Big/ T\left(\sum_{\tau=1}^{T}\tau^2 - \left(\frac{1}{T}\sum_{\tau=1}^{T}\tau\right)^2\right)\right). \tag{35}$$

For undetermined p^m, the trend constant estimate is t-distributed with T−2 degrees of freedom. Therefore a (100−2P)% catastrophic confidence interval is given by

$$\left|\hat{B}^m - B^m\right| < t_c\sqrt{\hat{\rho}^m \sum_{\tau=1}^{T}\tau^2 \Big/ T\left(\sum_{\tau=1}^{T}\tau^2 - \left(\frac{1}{T}\sum_{\tau=1}^{T}\tau\right)^2\right)} \tag{36}$$

where P is a predetermined number to provide a desired confidence level. For example, a "P" value of 0.5 provides a confidence level of 99%. The predicted value of the distance, $D^m(\tau_0)=d(P_i^m, P_{i-\tau_0}^m)$, of a future partial discharge measurement p_i^m and the partial discharge measurement τ_0 steps behind it may be obtained from the trend line mode defined in equation (37):

$$\hat{D}^m(\tau_0)=\hat{A}^m\tau_0+\hat{B}^m. \tag{37}$$

If the variance of the process is known, the predicted value $\hat{D}^m(\tau_0)$ has a Gaussian distribution based on the formula in equation (38):

$$N\left(A^m\tau_0 + B^m, \rho^m\left\{\frac{1}{T} + \left(\tau_0 - \frac{1}{T}\sum_{\tau=1}^{T}\tau\right)^2 \Big/ \left(\sum_{\tau=1}^{T}\tau^2 - \left(\frac{1}{T}\sum_{\tau=1}^{T}\tau\right)^2\right)\right\}\right). \tag{38}$$

The predicted distance $\hat{D}^m(\tau_0)$ is t-distributed with T−2 degrees of freedom. Therefore the (100−2P)% confidence interval is given by

$$\left|\hat{D}^m(\tau_0) - D^m(\tau_0)\right| < t_c\sqrt{\hat{\rho}^m\left\{\frac{1}{T} + \left(\tau_0 - \frac{1}{T}\sum_{\tau=1}^{T}\tau\right)^2 \Big/ \left(\sum_{\tau=1}^{T}\tau^2 - \left(\frac{1}{T}\sum_{\tau=1}^{T}\tau\right)^2\right)\right\}} \tag{39}$$

Steps of the condition monitoring method are summarized as follows:

Step 1. Collect "Z" number of PD measurements and compute the distance of the most recent measurements to the τ step back partial discharge measurements, for each τ≦T where T is a value less than the number of measurements "Z."

Step 2. Forming a parametric trend relationship between the distances $D^m(\tau)$ and $D^w(\tau)$ and the time lag τ as illustrated in equations (26) through (29).

Step 3. Estimate the trend parameters as described in equations (33)–(36).

Step 4. Predict the next step of the future partial discharge activity as described in equation (37).

Step 5. Collect the next partial discharge measurement and compute the one step difference as described in equation (24) or (25).

Step 6. Compare the difference between the actual one step distance and the predicted distance with the value described in equation (39). If the actual one step distance is within allowable limits, update the trend parameter estima-

tion using the new measurement. Otherwise tag the measurements as a potential indication of the change in the condition of the insulation system. Repeat the prediction-comparison procedure for a number of times to assure an indication of change in the condition of the insulation system.

Step 7. Estimate new trend parameters for the new condition.

Step 8. Compare the new "trend rate" and constant with the previous "trend rate" and constant using the equations (34) and (36). If both are beyond the values described in equations (34) and (36), generate an indication denoting a potential pre-catastrophic condition of the measured insulation. Repeat the observations to verify the pre-catastrophic condition.

The condition monitoring process as defined in this section provides a method of monitoring high voltage electrical equipment to predict immanent catastrophic insulation failures. This invention thus obviates the need for condition-based monitoring of electrical insulation within high voltage equipment in service.

It will be apparent to those skilled in the art that, while the invention has been illustrated and described herein in accordance with the patent statutes, modifications and changes may be made in the disclosed embodiments without departing from the true spirit and scope of the invention. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the invention.

What is claimed is:

1. A statistical method of condition monitoring the insulation in high voltage equipment utilizing partial discharge measurements, comprising the following steps:

calculating at least one analysis feature of undetermined insulation based on a plurality of partial discharge measurements;

estimating trend parameters of said at least one analysis feature based on a linear trend line generated based on a plurality of previously determined ones of said analysis feature;

predicting an estimated analysis feature statistical distance of said analysis feature with respect to said trend line;

comparing said predicted analysis feature statistical distance with an actual analysis feature statistical distance so as to determine whether the condition of said undetermined insulation has changed.

2. The statistical method of condition monitoring as recited in claim **1** further comprising the steps of:

conditioning said partial discharge measurements for computer manipulation;

generating a plurality of partial discharge measurements from said high voltage equipment with undetermined insulation so as to calculate at least one said analysis feature of said undetermined insulation.

3. The statistical method of condition monitoring as recited in claim **2** further comprising the steps of estimating said trend parameters of said analysis features based on a parabolic trend line generated in correspondence with previously determined ones of said analysis features.

4. The statistical method of condition monitoring as recited in claim **1** further comprising the steps of estimating said trend parameters of said analysis features based on a parabolic trend line generated in correspondence with previously determined ones of said analysis features.

5. The statistical method of condition monitoring as recited in claim **1** further comprising the steps of estimating said trend parameters of said analysis features based on a log linear-trend line generated in correspondence with previously determined ones of said analysis features.

6. The statistical method of condition monitoring as recited in claim **1** further comprising the step of generating a change in condition signal when the measured statistical distance is outside a normal operating limit.

7. The statistical method of condition monitoring as recited in claim **1** further comprising the steps of:

synchronizing the phase of said partial discharge measurements over a single cycle of a supply voltage;

normalizing the partial discharge magnitude of said partial discharge measurements; and

removing excitation noise from said partial discharge measurements.

8. The statistical method of condition monitoring as recited in claim **7** further comprising the step of taking the logarithm of said partial discharge magnitudes to minimize the low magnitude high partial discharge event count.

9. The statistical method of condition monitoring as recited in claim **8** further comprising the steps of comparing the latest calculated said trend parameters with the previously calculated ones of said trend parameters.

10. The statistical method of condition monitoring as recited in claim **9** further comprising the step of generating a pre-catastrophic insulation signal when the difference in a plurality of previous ones of said trend parameters and current said trend parameters are greater than a catastrophic confidence interval value.

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